

♠ Size of Variables

Understanding memory usage & type sizes in C#

In **Part 1** of this chapter, we explored what variables are and their different types. In **Part 2**, we go deeper — learning how to measure the **memory size** of each variable type and how to **convert** between them. The **sizeof()** operator is our main tool for this job.

01

Size of Integer (int)

An **int** stores whole numbers. The **sizeof(int)** operator tells us exactly how many bytes it occupies in memory.

● C# — Integer Size

```
using System;
class Program
{
    static unsafe void Main(String[] args)
    {
        int number = 20;
        Console.WriteLine(sizeof(int) + " bytes");
    }
}
```

OUTPUT

sizeof(int) →

4 bytes

02

Size of Double (decimal)

A **double** stores floating-point (decimal) numbers with high precision. It uses **8 bytes** — twice the size of an int.

● C# — Double Size

```
using System;
class Program
{
    static unsafe void Main(String[] args)
    {
        double number = 30.5;
        Console.WriteLine(sizeof(double) + " bytes");
    }
}
```

OUTPUT

sizeof(double) →

8 bytes

03

Size of Character (char)

A **char** holds a single Unicode character. Unlike C/C++, C# uses **UTF-16 encoding**, so every char takes **2 bytes**.

● C# — Char Size

```
using System;
class Program
{
    static unsafe void Main(String[] args)
    {
        char letter = 'A';
        Console.WriteLine(sizeof(char) + " bytes");
    }
}
```

OUTPUT

sizeof(char) →

2 bytes

04

Size of String

A **string** is a sequence of chars. C# does not have a direct **sizeof(string)** because strings are reference types with variable length. We calculate size manually:

● C# — String Size

```
using System;
class Program
{
    static unsafe void Main(String[] args)
    {
        string name = "Console";
        Console.WriteLine(name.Length * sizeof(char) + " bytes");
    }
}
```

OUTPUT	name.Length * sizeof(char) → "Console" = 7 chars →	14 bytes
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Step 1	name.Length	Counts characters in the string →	7
Step 2	sizeof(char)	Size of one char in bytes →	2
Step 3	7 × 2	Total memory used →	14 bytes

Quick Reference — Variable Sizes

Type	Keyword	Size	Example Value	sizeof()
Integer	int	4 bytes	20	sizeof()
Decimal	double	8 bytes	30.5	sizeof()
Character	char	2 bytes	'A'	sizeof()
String*	string	Variable	"Console"→14	Length × sizeof(char)

* String size = number of characters × 2 (because each char is 2 bytes in C# UTF-16)